A Three-Chamber Dynamic Release Device for Simultaneous Drug-Release Kinetics and *In Situ* Infection Challenge of Drug-Eluting Ultra-High-Molecular-Weight Polyethylene (UHMWPE) Joint Implants

Shorter title idea: DRD-3: A Three-Chamber Platform for Drug Release and Infection Challenge Testing of Drug-Eluting UHMWPE Joint Implants

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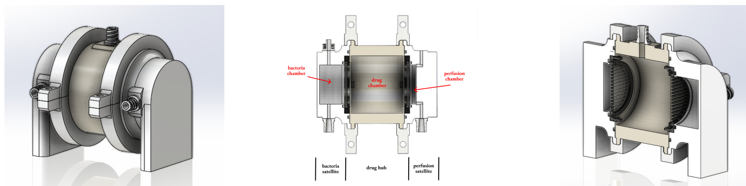

(A) as-modeled assembly **(B)** sectioned chambers **(C)** opposite-face membranes

Figure 1. The DRD-3 three-chamber device. **(A)** As-modeled assembly with the two-piece C-ring wrappers that load each joint. **(B)** Top-down section through the three chambers: a central drug hub (the joint space, holding the UHMWPE implant strip) is flanked on opposite faces by a perfusion satellite and a sealed bacteria satellite. **(C)** Isometric cut showing the two membranes in modular cartridges on opposite faces; placing them on opposite faces rather than in series keeps biofilm on the sacrificial 0.2 µm barrier from reaching the transport membrane that sets the kinetics. Sealing uses three identical O-ring pairs (2.0 mm cord, 25% compression).

Introduction. Periprosthetic joint infection (PJI) remains one of the most serious complications after total joint arthroplasty. An otherwise successful reconstruction can be converted into a prolonged course of revision surgery and antibiotic treatment, leading to impaired function and increased patient morbidity.[1,2] Although systemic antibiotics are routinely used in PJI treatment, achieving sustained therapeutic concentrations inside the joint is difficult because intraarticular drug availability is limited by local transport barriers, synovial fluid interactions, and the need to avoid toxic serum concentrations.[1,3] Local antibiotic delivery from drug-eluting implants is therefore attractive because it can concentrate drug at the site of infection while reducing systemic exposure.

Ultra-high-molecular-weight polyethylene (UHMWPE) is the material of choice for load-bearing total joint implants, and antibiotic-eluting UHMWPE is being developed as a functional spacer or implant material for PJI treatment.[1,2] Translation of these materials requires in-vitro models that capture the release, transport, and clearance conditions governing drug concentration inside the joint. The Dynamic Release Device (DRD) was recently developed to address this need by combining a drug chamber representing the joint space, a diffusion membrane representing synovial transport, and a perfused chamber representing clearance.[3] However, the DRD evaluates drug transport without a live infection challenge, as the current design would mean adding bacteria directly to the drug chamber; this would allow biofilm to foul the transport membrane, thus altering the permeability that defines the measured release kinetics. In this study, we designed a three-chamber DRD-3 to preserve the DRD transport measurement while adding a physically isolated *Staphylococcus aureus* challenge in the same experiment, furthering our understanding of intrarticular dynamics by simualting the infection-based scenario where a drug-eluding antiobiotic would be most useful.

Methods. We designed a clean-sheet, three-chamber device (the DRD-3) in CAD (Fig. 1A). It targets a tunable fluid volume of 4 to 20 mL spanning healthy to infected knee synovial volumes. A central drug hub represents the joint space, and therefore holds the UHMWPE implant strip. The hub is flanked on opposite faces by two satellites: a perfusion satellite providing continuous fresh-media clearance through validated transport physics and a bacteria satellite housing a sealed culture (Fig. 1B). Ergo, the two membranes themselves sit on opposite faces of the hub (Fig. 1C). Each membrane is sealed off-device inside a modular lid-and-base *cartridge*; a single tapered tooth on the cartridge lid (tapering from 2.0 to 1.0 mm) both captures the membrane's base and retains the finished cartridge in its satellite, while parallel protective bars shield the membrane from the UHMWPE strip without restricting diffusion. Every sealing interface uses one identical static O-ring specification (2.0 mm cord, 1.5 mm gland, 0.5 mm stand-proud giving 25% compression), so the entire three-chamber device seals on three identical O-ring pairs. Each satellite is drawn to the hub by a two-piece C-ring wrapper whose U-channel applies uniform radial load around the joint. Bacterial isolation is enforced by a sacrificial 0.2 µm microfilter on the bacteria-facing membrane that passes drug freely but physically blocks *S. aureus* (0.5 to 1 µm) from migrating toward the transport membrane; the culture is accessed through a self-sealing septum for inoculation and colony-forming-unit (CFU) sampling without breaching sterility. Structural parts are stereolithography (SLA) resins (Accura-60-class chassis; Somos 9120 for the compliant inner cartridge parts) with medical-grade silicone/EPDM (USP Class VI) O-rings for watertight capacity and sealing. A three-phase validation protocol is defined: Phase 0, a dye-perfusion leak test at 64 mL/h for 24 h (acceptance below 2% volume loss per 24 h); Phase 1, reproduction of the published release kinetics across flow rate (2 to 64 mL/h), temperature (room temperature versus 37 °C), and membrane thickness; and Phase 2, introduction of *S. aureus* in the bacteria satellite with CFU sampling over 24 h to confirm zero bacterial migration to the perfusion side. As a non-clinical benchtop instrument, this work involves no human or animal subjects, and sex as a biological variable is not applicable at this stage.

Results. A complete three-chamber iteration of the dynamic release device is designed, manufactured, and validated. The design consolidates all sealing to three identical O-ring pairs and a single tapered-tooth feature that performs both membrane capture and cartridge retention; such a design replaces the earlier screw-based and multi-bolt, point-loaded clamp with one continuous C-ring wrapper per satellite (four wrappers, one bolt per side) that distributes radial load uniformly around each joint. The 0.2 µm isolation barrier is sized well below the *S. aureus* cell diameter, and the drug hub is parameterized so hub depth can be tuned to a target volume. Experimental validation is in progress: Phase 0 leak loss **[value % / 24 h]**; Phase 1 agreement of DRD-3 half-life trends with the published baseline **[result]**; Phase 2 bacterial count on the perfusion side **[CFU; expected 0]**.

Discussion. The defining feature of the DRD-3 is the radial, opposite-face, two-membrane architecture, which not only introduces an infection challenge to the system, but decouples it from the transport measurements. The modular design targets the reproducibility and assembly failure modes of press-fit Stereolithography (SLA) stacks: the membrane is sealed as a bench-built module, the seal is an elastomer rather than a flexing printed feature, and the wrapper clamp removes press-fit interfaces from the load path. A limitation of the design include the fact SLA parts must be fully post-cured and washed before live-culture runs to avoid residual-monomer antimicrobial bias. The mapping to anatomy is also deliberately partial: the drug hub represents the intraarticular space and the perfusion satellite the joint's clearance pathway, but the isolated bacterial compartment is an engineering abstraction rather than an anatomical analog (in vivo, the infection resides within the joint space itself). Such a method was adopted so that biofilm cannot foul the transport membrane that defines the kinetics. Within the framework of our objective, the device is built so that, once it reproduces the published transport baseline, it can add a live antimicrobial-efficacy readout without compromising that measurement.

Significance. This platform would allow drug-eluting UHMWPE bearing materials to be screened for both intraarticular release kinetics and live antimicrobial efficacy on a single, compact benchtop device, directly supporting the development of local-delivery strategies and materials innovations against periprosthetic joint infection.

Acknowledgements. Will go here. **[funding sources will go here when I talk to Dr. Asik]**

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